

Blockchain 101

Your first smart contract

UMons

What is a blockchain?



A **public database** that anyone can read.
Append-only. Signed. Unstoppable.

Smart contracts = programs that live on it.

The 4 words you need

Wallet

a key pair
= *you*

Contract

a program
+ state

Transaction

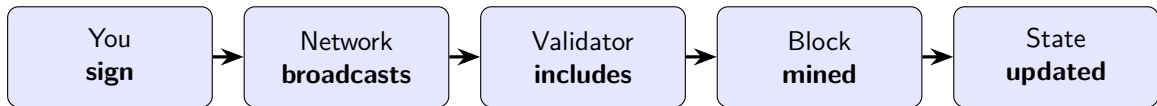
a signed
instruction

Gas

the price of
a write

Reading is **free**. Writing costs **gas**.

Life of a transaction



Takes a few seconds on Arbitrum. Costs a fraction of a cent.

*Once mined, the change is **permanent** and visible to the whole world.*

What we will build

A contract called SimpleStorage

value

owner

lastValueBy[address]

set(uint256 x)

increment()

reset() *only owner*

event ValueChanged(...)

- Stores a number
- Anyone can change it
- Tracks who wrote what
- Only the deployer can reset

What's an event?

A log line the contract emits.

It does *not* change state, but it is **saved on-chain** and anyone can subscribe to it — great for dApps that want to react in real time.

Today's goal



Your code, live on the public Internet.

2 ways to call your contract

1. Explorer

click buttons
on Arbiscan

2. Script

TypeScript,
the way a dApp does

Same contract. Two angles to understand it.

Read = free. Write = signed transaction + gas.

Resources

Docs `tanguyvans.github.io/blockchain-101`

Repo `github.com/tanguyvans/blockchain-101`

Network Arbitrum Sepolia (chain id 421614)

Explorer `sepolia.arbiscan.io`

Faucet `faucet.quicknode.com/arbitrum/sepolia`

Need test ETH? Email me your public address at
`tanguy.vansnick@umons.ac.be` and I will send you faucet funds.

Let's build.